MONITORING AND LOGGING WITH CLOUDWATCH

PRACTICAL-1

CLOUDWATCH AGENT (Monitor Memory & Disk usage (Custom Metrics))

1.Create EC2 Instance

1. name ------ cw-agent-demo
2. machine --- ubuntu
3. sg -------- http/ssh/
4. create a instance

2.Attach IAM Role to Instance

1.Role_name: (CloudWatchAgentServerPolicy) or administrator access

3.Connect to Instance

1. sudo apt update
2. wget https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-cloudwatch-agent.deb
3. sudo dpkg -i amazon-cloudwatch-agent.deb
4. sudo apt-get install -f -y

4.configure agent

1.sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard

cwagent → Enter
StatsD → NO
CollectD → NO
Metrics → YES
CPU → YES
Memory → YES
Disk → YES
Interval → 60
Logs → NO
X-Ray → NO
SSM → NO
Default → Basic

Final → YES

5.Apply config

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json \
-s
```

6.Start and Enable Agent

1. sudo systemctl start amazon-cloudwatch-agent
2. sudo systemctl enable amazon-cloudwatch-agent
3. sudo systemctl status amazon-cloudwatch-agent

7.Verify Metrics

go to cloudwatch → metrics → custom namespace → CWAgent

8.create a SNS topic

1. create a topic
2. type ------- standard
3. name ----- memory-alert
4. create a topic

9.create a subscription in SNS

1. create a subscription
2. topic arn ----> memory-alert
3. protocol -----> email
4. create subscription and go to ur gmail and confirm subscription

10.create alarm

1. cloudwatch ----> alarms ----> create alarm
2. metric ----> custom namespace ----> CWAgent
3. InstanceId ----> select instance
4. metric ----> mem_used_percent
5. period ----> 1 minute
6. threshold -------> greater/equal than 50
7. select sns -------> memory-alert
8. alarm name ----> memory-alarm

9. create alarm

11.Connet Instance And Add Stress

1. sudo apt install stress -y
2. stress --vm 1 --vm-bytes 200M --timeout 300s

12.Check The Metrics Utilization Status

click on metrics -----> custom namespace ----> CWAgent ----> mem_used_percent ----> select number

12.output

1.memory increase ----> alarm in alarm state -----> email received

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PRACTICAL-2

CLOUDWATCH LOGS PRACTICAL (USING CLOUDWATCH AGENT)

1.Create EC2 Instance

1. name ------ cw-agent-demo
2. machine --- ubuntu
3. sg -------- http/ssh/
4. create a instance

2.Attach IAM Role to Instance

1.Role_name: (CloudWatchFullAccess) or administrator access

3.Connect to Instance

1. sudo apt update
2. sudo apt install nginx -y
3. sudo systemctl start nginx
4. sudo systemctl enable nginx
5. curl localhost

4.Add permissions to the logs

1. sudo chmod 644 /var/log/nginx/access.log

2. sudo chmod 644 /var/log/nginx/error.log

5.Install Agent on ubuntu

1. wget https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-cloudwatch-agent.deb
2. sudo dpkg -i amazon-cloudwatch-agent.deb
3. sudo apt-get install -f -y

4.Configure Agent

1.sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard

{ cwagent → Enter
StatsD → NO
CollectD → NO
Metrics → YES
CPU → YES
Memory → YES
Disk → YES
Interval → 60
Logs → Yes

Log file path:

1./var/log/nginx/access.log
2.Log group:
nginx-access
3.Add more logs → YES

Log file path:

1./var/log/nginx/error.log
2.Log group:
nginx-error

X-Ray → NO
SSM → NO
Default → Basic
Final → YES }

5.Apply config

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json \
-s
```

6.Start and Enable Agent

1.sudo systemctl start amazon-cloudwatch-agent
2.sudo systemctl enable amazon-cloudwatch-agent
3.sudo systemctl status amazon-cloudwatch-agent

7.Generate Logs

1. curl http://localhost
2. curl http://localhost
3. curl http://localhost

8.<public_ip> hit on web

9.Verify

1.Go to cloudwatch ----> logs -----> logs group -----> 1.nginx-access
2.nginx-error
